

Customer360 Data Integration Platform

Project Overview

Customer360 is an ETL pipeline developed to combine customer information from multiple airline source systems into a single customer-level view.

The pipeline integrates:

- Customer profile data
- Flight booking data
- Loyalty information
- Support ticket information

The system reads source CSV files, validates data quality, cleans and transforms records, calculates customer metrics, and generates a consolidated Customer360 output.

Key implementation features:

- Modular ETL pipeline design
- Data validation and quality checks
- Invalid record tracking
- Missing and incorrect data handling
- Customer-level aggregation
- Automated output generation

Technology Used

Programming Language:

- Python

Libraries:

- Pandas → Data processing and transformation
- Regex → Email validation
- Logging → Pipeline monitoring
- Faker → Random source data generation

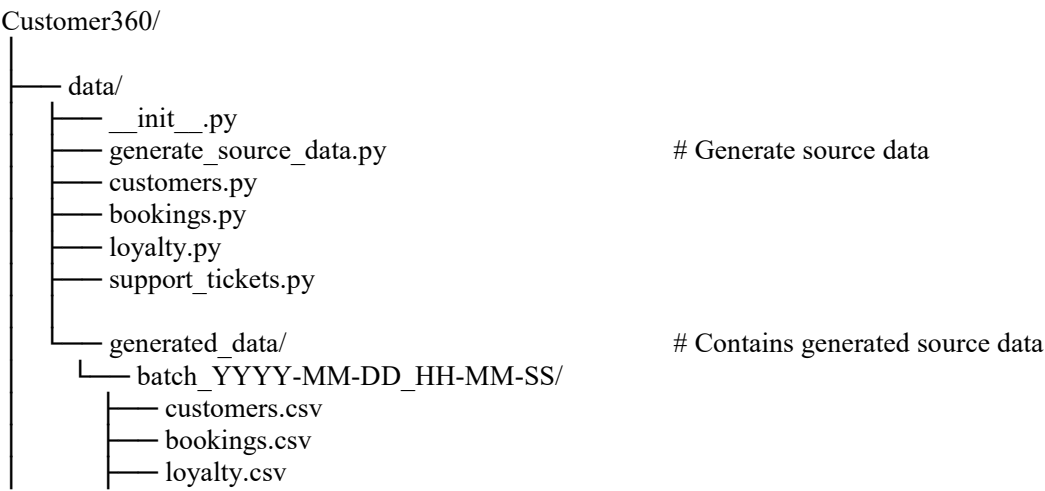
Input Format:

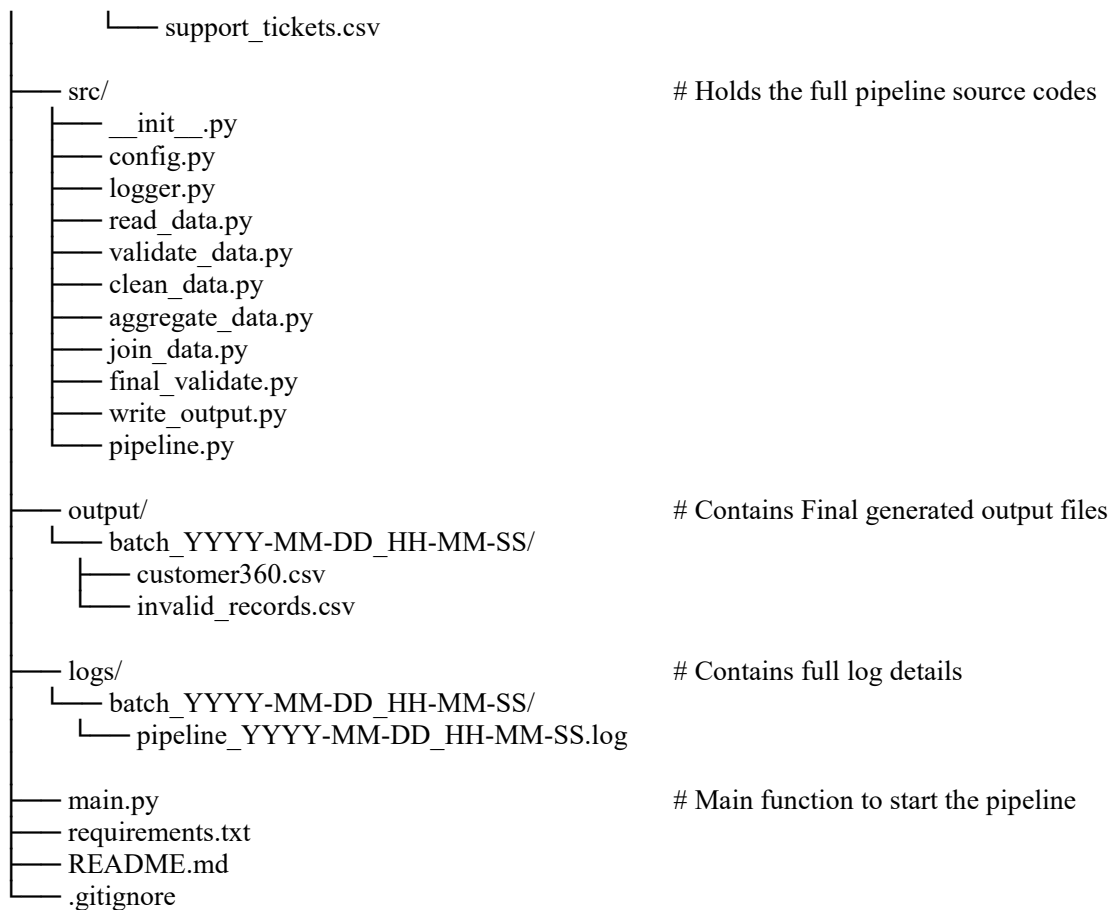
- CSV files

Output Format:

- CSV files

Project Folder Structure





Input Data Sources and File Details

customers.csv

Column Name	Description
customer_id	Unique identifier for each customer
first_name	Customer first name
last_name	Customer last name
email	Customer email address
phone	Customer contact number
country	Customer country

bookings.csv

Column Name	Description
booking_id	Unique identifier for each booking
customer_id	Customer reference identifier
flight	Flight number/details
booking_date	Date of booking
amount	Booking transaction amount
status	Booking status (Completed / Cancelled)

loyalty.csv

Column Name	Description
customer_id	Customer reference identifier
membership_type	Type of membership
loyalty_points	Reward points earned by customer
loyalty_tier	Customer loyalty category

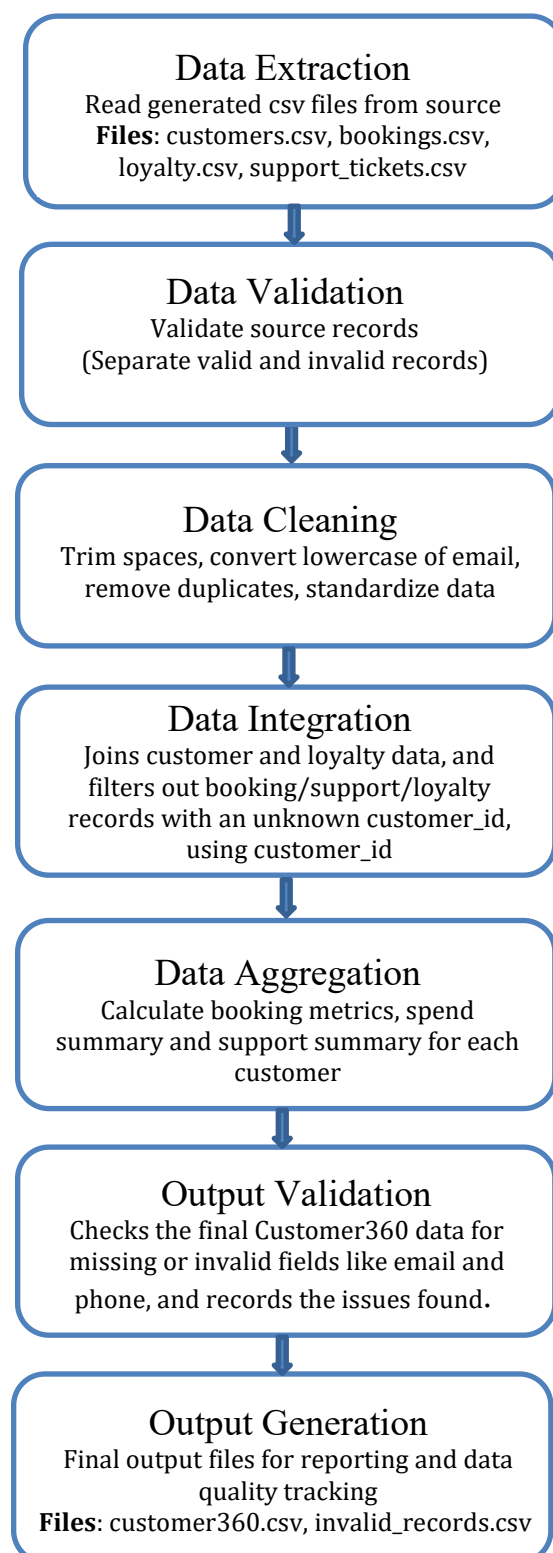
support_tickets.csv

Column Name	Description
ticket_id	Unique identifier for support ticket
customer_id	Customer reference identifier
issue_type	Type of customer issue
created_date	Ticket creation date
status	Ticket status (Open / Closed)

ETL Pipeline

- generate_source_data.py – Generates random source data and creates timestamped batches.
- read_data.py – Reads the four source CSV files.
- validate_data.py – Checks source data quality and identifies invalid records.

- `clean_data.py` – Cleans and standardizes the source data.
- `aggregate_data.py` – Calculates customer-level booking and support metrics.
- `join_data.py` – Combines customer data with aggregated information.
- `final_validate.py` – Performs final Customer360 data validation.
- `write_output.py` – Creates the final Customer360 and invalid-record files.
- `logger.py` – Creates and manages timestamped pipeline logs.
- `config.py` – Stores common project paths, files, and settings.
- `pipeline.py` – Controls the complete ETL pipeline flow.
- `main.py` – Starts source generation and runs the pipeline.



Data Quality Handling

The system performs data-quality checks to identify and handle invalid or inconsistent source records.

- **Missing Customer ID:** Records without `customer_id` are rejected and stored in `invalid_records.csv`.
- **Invalid Email:** Invalid email values are replaced with NULL and recorded in the validation report.
- **Invalid Phone Number:** Invalid phone values are replaced with NULL while retaining the customer record.
- **Missing Data:** Missing optional fields are retained as NULL in the final Customer360 output.
- **Duplicate Records:** Duplicate customer records are identified and handled during the cleaning process.
- **Invalid Relationships:** Booking, loyalty, and support records referencing unknown `customer_id` values are rejected during data integration.
- **Error Tracking:** Validation issues are recorded with the source system, record ID, field, invalid value, reason, and action.

Data Transformation and Business Logic

The pipeline transforms validated source data into a consolidated customer-level dataset.

- **Customer Data Cleaning:** Standardizes text values and removes unwanted spaces.
- **Booking Metrics:** Calculates total bookings, completed bookings, cancelled bookings, total spend, average booking value, and last booking date.
- **Average Booking Value:** Calculated from the available booking metrics and returned as 0 when no applicable booking data exists.
- **Support Analysis:** Calculates total, open, and closed support tickets.
- **Loyalty Integration:** Adds loyalty points and loyalty tier information to the customer profile.
- **Customer-Level Integration:** Combines all processed information using `customer_id`.

Output Files

The pipeline generates two output files for each pipeline batch.

1. `customer360.csv`

- Contains the consolidated customer-level view.
- Includes customer profile, booking metrics, loyalty information, and support details.
- Acts as the final business-ready dataset.
- Missing values are represented as NULL.
- Count-based fields are stored as whole numbers.
- `last_booking_date` is stored as a date without a time component.

Example location:

```
output/  
└─ batch_2026-08-24_15-49-05/  
    └─ customer360.csv
```

2. invalid_records.csv

- Contains validation issues identified during processing.
- Stores source system, customer/record ID, field, invalid value, reason, and action.
- Helps track and review data-quality problems.

Example:

```
output/  
└─ batch_2026-08-24_15-49-05/  
   └─ invalid_records.csv
```

Logging and Error Handling

The pipeline uses centralized logging and exception handling to monitor every execution.

- Creates a separate timestamped log file for each pipeline run.
- Logs data generation, reading, validation, cleaning, aggregation, joining, and output stages.
- Stores logs inside the corresponding batch log folder.
- Records validation issues in invalid_records.csv.
- Captures pipeline errors and exceptions in the log file.
- Displays only the pipeline status and log-file location in the terminal.
- Prevents silent failures and supports easy troubleshooting.

Example:

```
logs/  
└─ batch_2026-08-24_15-49-05/  
   └─ pipeline_2026-08-24_15-49-05.log
```

Final Outcome

The Customer360 pipeline produces a unified customer-level dataset by integrating customer, booking, loyalty, and support data.

- Generates business-ready customer360.csv.
- Generates invalid_records.csv for data-quality issues.
- Stores source data, outputs, and logs in timestamped batch folders.
- Maintains complete batch-wise traceability from source data to final output.
- Provides detailed logs for monitoring and troubleshooting.